MIRA: Mutual Information guided calibration for Reliable Test-Time Adaptation

Hyeongyu Kim[1,4*] **Mingyeong Jang**[1*] **Youngjun Song**[1] **GeonHui Han**[1] **Dosik Hwang**[1,2,3†]
[1]School of Electrical and Electronic Engineering, Yonsei University
[2]Department of Radiology and Center for Clinical Imaging Data Science,
College of Medicine, Yonsei University
[3]Artificial Intelligence and Robotics Institute, Korea Institute of Science and Technology
[4]Semiconductor Development Group, Hyundai Motor Company
{lion4309, mini10211, dosik.hwang}@yonsei.ac.kr

Abstract

Well-calibrated confidence estimates are important for reliable decision-making when models encounter distribution shift. However, entropy-based test-time adaptation can reinforce confident errors even when accuracy improves, without necessarily yielding well-calibrated predictions. We propose MIRA, **M**utual **I**nformation guided calibration for **R**eliable Test-Time **A**daptation, a source-free online method that probes prediction reliability with Gaussian perturbations of the final classifier. MIRA computes backbone features once per target batch and reuses them across classifier samples and candidate perturbation scales. A mutual-information (MI) criterion and a prediction-agreement constraint select a batch-specific probing radius without target labels or posterior fitting. The resulting local disagreement and confidence define detached sample weights for a stochastic entropy objective. The stochastic objective supplies the adaptation signal, while reliability weighting reduces the influence of perturbation-sensitive samples. Experiments with CNNs, a vision transformer, and CLIP show improvements in the accuracy–calibration trade-off. Further evaluations cover continual and dynamic adaptation, stronger representation shifts, and computational time constraints. MIRA provides a reliability module whose accuracy gains depend on the base adaptation method; its MI is an operational stability signal, rather than calibrated Bayesian posterior uncertainty.

1 Introduction

Distribution shift degrades both the accuracy and calibration of deployed neural networks [9, 19]. Test-time adaptation (TTA) seeks to recover predictive performance by updating a source-trained model using unlabeled target samples during inference [21, 23, 25, 16]. A common strategy is to minimize prediction entropy, encouraging the model to make confident predictions. However, increased confidence does not necessarily reflect improved correctness: entropy minimization can reinforce confident mistakes and worsen calibration [2, 28]. Consequently, accuracy improvements alone do not establish that the adapted model's confidence is trustworthy.

Reliable TTA therefore requires considering accuracy and calibration together. The goal is to make confidence reflect prediction correctness as the model adapts. Both reducing unsupported confidence and increasing confidence when predictions become more accurate can contribute to this goal. We assess this accuracy-calibration trade-off using expected calibration error (ECE), which measures confidence–accuracy mismatch, and negative log-likelihood (NLL), which penalizes confident errors

*These authors contributed equally to this work.
†Corresponding author.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

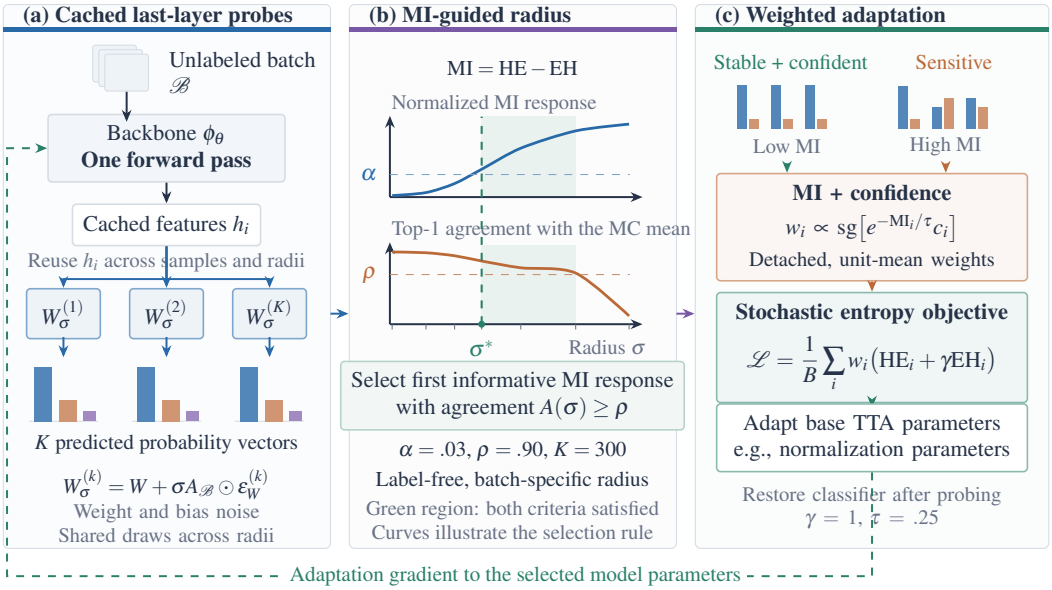


Figure 1: **MIRA overview.** (a) A single backbone forward supplies cached features for Gaussian last-layer probes. (b) The earliest informative MI response with high top-1 agreement determines a batch-specific radius. (c) Detached MI–confidence weights redistribute sample contributions to the core stochastic entropy objective; gradients update the selected adaptation parameters. HE is entropy of the mean prediction and EH is mean predictive entropy. Specialized optimizers use the integration in Section 4. Curves and probability bars are schematic illustrations.

across the full predictive distribution. Better-calibrated confidence can support selective prediction, although application-specific decision guarantees require separate validation.

To address these reliability concerns, existing TTA methods filter samples, estimate adaptation quality, or introduce probabilistic modeling [17, 18, 14, 22, 30, 4, 31]. Their computational and information requirements vary, with some approaches relying on repeated stochastic inference or source-side statistics. Last-layer uncertainty estimators offer an efficient alternative by reusing backbone features [10, 5, 29]. However, constructing a posterior approximation can introduce additional fitting steps, such as data-dependent curvature estimation or prior-precision tuning [5, 29]. These requirements can be difficult to accommodate when deployment provides only a pretrained model and unlabeled target batches, especially if the approximation must be refreshed as the model adapts. For reliability-weighted adaptation, these constraints motivate directly probing the current model's prediction stability through controlled classifier perturbations. We therefore ask whether such perturbations can provide a useful, label-free signal for weighting adaptation updates without fitting a posterior.

To this end, we propose **MIRA**, **M**utual **I**nformation guided calibration for **R**eliable Test-Time **A**daptation, a source-free online method for improving the accuracy-calibration trade-off. We consider a fixed label space under covariate shift [20], with no target labels available for adaptation or scale selection. Starting from a deterministic maximum a posteriori (MAP) model, the method computes backbone features once per target batch and reuses them across Gaussian samples of the final classifier and candidate perturbation scales. A non-trivial mutual-information (MI) response and high class agreement guide the selection of a batch-specific probing radius. The resulting MI measures local perturbation-induced disagreement, not a calibrated probability of error. At the selected radius, a stochastic entropy objective supplies the adaptation signal, while detached weights combining MI and confidence reduce the contribution of perturbation-sensitive samples. Figure 1 illustrates the three operations: probing the classifier, selecting a radius, and adapting with detached sample weights. Controlled ablations assess the contributions of the stochastic objective and reliability weights separately.

Our contributions are as follows:

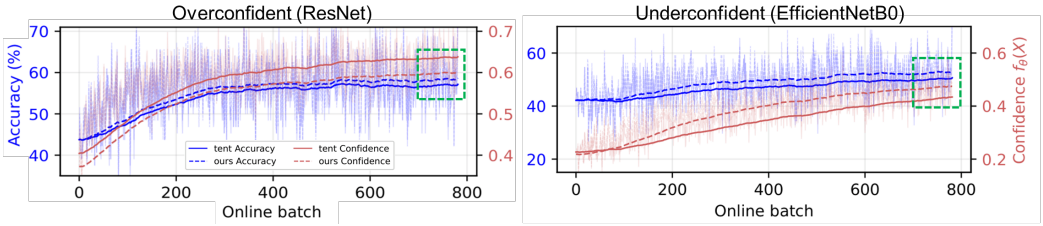


Figure 2: Accuracy and confidence under adaptation. The ResNet example illustrates overconfidence and the EfficientNet example illustrates underconfidence. MIRA can reduce or increase confidence as accuracy improves; the desired outcome is better prediction quality and calibration, rather than uniformly lower confidence.

- **Online reliability probing without target labels.** We introduce a last-layer Gaussian probe with batch-specific scale selection guided by MI and prediction agreement, reusing cached backbone features without posterior fitting.
- **Stochastic adaptation guided by sample reliability.** We combine a stochastic entropy objective with detached MI–confidence weights to reduce the contribution of perturbation-sensitive samples. Controlled ablations distinguish the effects of the objective and reliability weighting.
- **Evaluation of accuracy, calibration, and computational cost.** We evaluate CNN, ViT, and CLIP models under multiple online adaptation protocols, characterizing method-dependent accuracy gains, calibration improvements, and computational trade-offs.

2 Background

2.1 Test-time adaptation and uncertainty

TTA methods update normalization statistics or parameters, minimize prediction entropy, enforce consistency, or filter unreliable samples [15, 23, 17, 18, 24, 27]. For a target batch of size B, deterministic entropy minimization uses

$$\mathcal{L}_{\text{ent}} = B^{-1} \sum_{i=1}^{B} H(p_\theta(\cdot \mid x_i)), \qquad H(p) = -\sum_c p_c \log p_c. \tag{1}$$

Low entropy alone cannot distinguish a correct prediction from an overconfident error. EATA and SAR control unreliable updates [17, 18]; EATA-C explicitly addresses calibration [22]; AETTA estimates accuracy with dropout disagreement [14]; SSA uses steady-state adaptation [13]; and BACS and PETAL provide Bayesian or probabilistic formulations [30, 4]. MIRA supplies a complementary sample-reliability signal while retaining the base method's adaptation parameters.

2.2 Perturbation-based reliability probing

Monte Carlo dropout, ensembles, and Bayesian weight distributions estimate uncertainty through stochastic predictions [6, 12, 3]. Their disagreement can be informative under shift [19], but may require multiple models, repeated backbone inference, or fitting. Last-layer methods reuse cached features [10, 5, 29]. MIRA uses this computational structure to probe a local stochastic classifier family on the current target batch. MI captures changes in the full predictive distribution, while agreement tracks top-1 stability. Neither establishes correctness without additional assumptions.

3 Motivation and observation

Figure 2 contrasts the confidence trajectories of two architectures. In the overconfident case, increasing confidence without a corresponding accuracy gain widens their mismatch; in the underconfident case, indiscriminately lowering confidence would be counterproductive. These examples motivate

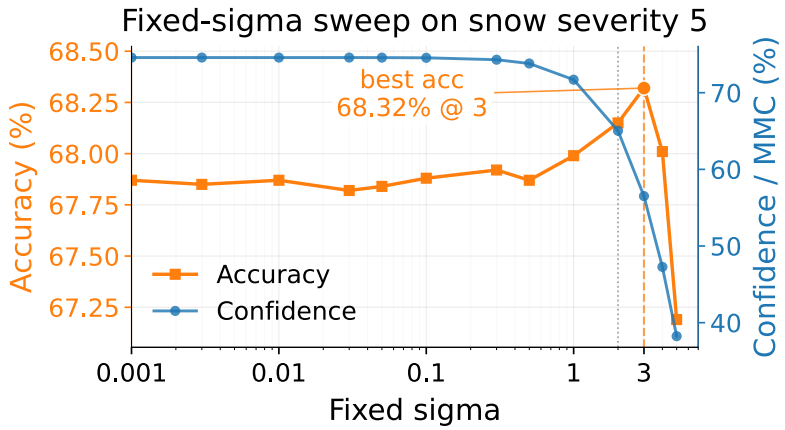


Figure 3: Fixed perturbation-radius sweep on ImageNet-C snow, severity 5. Accuracy and mean confidence are shown; an intermediate radius changes confidence without destroying predictive accuracy.

Table 1: Relative perturbation-scale comparison. Accuracy is in percent; σ denotes the reference radius in this diagnostic.

Scale	Acc. (%) ↑	MMC	ECE ↓
Source	60.52	0.7212	0.1160
TENT	67.86	0.7458	0.0678
MIRA: $0.01 \times \sigma$	67.87	0.7458	0.0677
MIRA: $1.0 \times \sigma$	67.99	0.7170	0.0381
MIRA: $3.0 \times \sigma$	68.32	0.5651	0.2734
MIRA: $5.0 \times \sigma$	67.19	0.3824	0.2895

interpreting MMC together with accuracy, ECE, and NLL, rather than treating smaller MMC as an independent success criterion.

We perturb the final classifier of a deployed model and sweep the perturbation radius on ImageNet-C snow at severity 5. Figure 3 is a real target-stream experiment, rather than a toy example. It illustrates three regimes summarized in Table 1. Small perturbations leave deterministic overconfidence largely unchanged. Moderate perturbations can lower unsupported confidence while preserving or improving accuracy during adaptation. Excessively large perturbations induce class changes and predictive collapse.

This motivates treating the radius as a local probing parameter rather than posterior variance. MIRA selects the earliest non-trivial MI response subject to high agreement, then uses the resulting disagreement to weight adaptation. The margin and small-perturbation derivations in Appendix A explain the distinct roles of these criteria. The all-corruption entropy-matched analysis in Appendix E provides broader empirical evidence: higher-MI batches tend to be less accurate even at similar predictive entropy. This association motivates reliability weighting but does not guarantee pseudo-label correctness.

4 Method

We write the deployed classifier as $z(x) = W\phi_\theta(x) + b$, with $W \in \mathbb{R}^{C\times D}$ and $p(y \mid x) = \mathrm{softmax}(z(x))$. MIRA updates the parameters selected by the underlying TTA method, such as normalization affine parameters; the classifier is fixed except for temporary reliability probes.

4.1 Last-layer Gaussian reliability probe

For each target batch $\mathcal{B} = \{x_i\}_{i=1}^{B}$, compute $h_i = \phi_\theta(x_i)$ once. At candidate radius σ, sample K classifiers:

$$W_\sigma^{(k)} = W + \sigma A_\mathcal{B} \odot \epsilon_W^{(k)}, \qquad b_\sigma^{(k)} = b + \sigma a_\mathcal{B} \odot \epsilon_b^{(k)}, \qquad \epsilon_W^{(k)}, \epsilon_b^{(k)} \sim \mathcal{N}(0, I). \tag{2}$$

The headline construction uses $s_0 = \sqrt{2/D}$ and $A_\mathcal{B}[c, d] = s_0 g_{\mathcal{B},d}$, $a_\mathcal{B}[c] = s_0$. Here $g_{\mathcal{B},d}$ is the batch mean absolute feature magnitude, divided by its mean over features and clipped to $[0.25, 4]$; it is broadcast across classes. Appendix D.4 gives the formulas and alternative construction rules. The same Gaussian draws are shared across candidate radii within a batch.

Define $p_{i,\sigma}^{(k)} = \mathrm{softmax}(W_\sigma^{(k)} h_i + b_\sigma^{(k)})$ and $\bar{p}_{i,\sigma} = K^{-1} \sum_k p_{i,\sigma}^{(k)}$. The predictive entropy, expected entropy, and MI are

$$\mathrm{HE}_{i,\sigma} = H(\bar{p}_{i,\sigma}), \qquad \mathrm{EH}_{i,\sigma} = K^{-1} \sum_k H(p_{i,\sigma}^{(k)}), \qquad \mathrm{MI}_{i,\sigma} = \mathrm{HE}_{i,\sigma} - \mathrm{EH}_{i,\sigma}. \tag{3}$$

MI measures disagreement within this perturbation family. Under a small, centered logit perturbation with covariance $\Sigma_{z,i}$, a second-order entropy expansion gives $\mathrm{MI}_{i,\sigma} \approx \frac{1}{2} \mathrm{tr}(J_i \Sigma_{z,i})$, where $J_i = \mathrm{diag}(p_i) - p_i p_i^\top$ is the softmax Jacobian at the reference prediction. MI therefore responds to changes in the full class-probability distribution, even before the winning class changes. This local approximation requires small perturbations and is not used as a correctness guarantee; Appendix A provides its assumptions and derivation. The top-1 agreement with the Monte Carlo mean is

$$a_{i,\sigma} = K^{-1} \sum_k \mathbf{1}\left[\arg\max_c p_{i,\sigma}^{(k)}(c) = \arg\max_c \bar{p}_{i,\sigma}(c)\right]. \tag{4}$$

4.2 MI-guided scale selection

Let $M(\sigma) = B^{-1} \sum_i \mathrm{MI}_{i,\sigma}$ and $A(\sigma) = B^{-1} \sum_i a_{i,\sigma}$. For an ascending candidate grid $\mathcal{G} = \{\sigma_1, \ldots, \sigma_L\}$, the discrete selection criterion is

$$\sigma^* = \min\left\{\sigma_\ell \in \mathcal{G} : \frac{M(\sigma_\ell) - M(\sigma_1)}{\max_j M(\sigma_j) - M(\sigma_1) + \epsilon} \geq \alpha, \quad A(\sigma_\ell) \geq \rho\right\}. \tag{5}$$

The normalized MI condition excludes a near-deterministic probe; agreement excludes excessive class changes. With $\rho = 0.90$, the batch-average empirical disagreement relative to the Monte Carlo mean is at most 10%. This is a stability constraint, not an error bound. For comparison, a fixed pairwise reference-class margin m_{ij} with Gaussian perturbation standard deviation σs_{ij} crosses zero with probability $\Phi(-m_{ij}/(\sigma s_{ij}))$. This explains the margin-stability rationale, while our implemented agreement uses the Monte Carlo mean as its reference class and directly measures empirical top-1 stability. The defaults are $\alpha = 0.03$, $\rho = 0.90$, $K = 300$, and batch size 64, without corruption-specific tuning. The finite grid and the reported interpolation setting are specified in Appendix D.5. The MI threshold α controls how informative a probe must be, whereas ρ limits class disagreement. Both are fixed across the headline experiments; the reported construction and sampling-budget ablations do not establish sensitivity to every threshold or grid choice.

4.3 Reliability-weighted adaptation

At the selected radius, abbreviate $m_i = \mathrm{MI}_{i,\sigma^*}$ and $c_i = \max_c \bar{p}_{i,\sigma^*}(c)$. For positive batch values, define $\mathrm{Norm}(v)_i = v_i / (B^{-1} \sum_j v_j)$. The detached reliability weight is

$$u_i = e^{-(m_i - \bar{m})/\tau} c_i, \qquad w_i = \mathrm{sg}[\mathrm{Norm}(u)_i], \qquad \bar{m} = B^{-1} \sum_i m_i, \tag{6}$$

where sg stops gradients through the weights and $\tau = 0.25$. This is equivalent to multiplying the individually normalized MI and confidence factors and renormalizing. The weights have unit batch mean and favor confident, perturbation-stable samples. They redistribute sample contributions; unit-mean weights alone do not imply identical gradient norms across objectives. Appendix B separately reports gradient-matched comparisons.

MIRA's stochastic adaptation loss is

$$\mathcal{L}_{\mathrm{MIRA}} = B^{-1} \sum_i w_i \big(\mathrm{HE}_{i,\sigma^*} + \gamma \mathrm{EH}_{i,\sigma^*}\big), \qquad \gamma = 1. \tag{7}$$

The entropy of the mean and the mean entropy have the same range $[0, \log C]$, motivating equal default coefficients. The unweighted stochastic objective supplies much of the adaptation gain; MI–confidence gating provides complementary sample-wise control. The gate is a reliability estimate, whereas Eq. (7) is the loss minimized to adapt the model. Neither guarantees correct pseudo-labels, and classifier-only probing cannot repair arbitrary feature collapse.

4.4 Cached sampling and integration

All sampled logits reuse the same features:

$$z_{i,\sigma}^{(k)} = W h_i + b + \sigma\big[(A_{\mathcal{B}} \odot \epsilon_W^{(k)}) h_i + a_{\mathcal{B}} \odot \epsilon_b^{(k)}\big]. \tag{8}$$

The extra Monte Carlo cost is confined to the classifier, rather than K backbone passes. Specialized optimizers need a compatible integration. For SAR, estimate and detach reliability weights at the unperturbed model, restore the classifier, then hold the weights fixed in both steps of the original sharpness-aware update. This separates stochastic reliability estimation from SAR's optimization perturbations and preserves its two-step update (Appendix D.6).

5 Experiments

5.1 Experimental setup

We evaluate CIFAR-10-C, CIFAR-100-C, and ImageNet-C [9], using corruption severity 5 unless stated otherwise. Headline plug-in results use ResNeXt on CIFAR-100-C and ResNet-50 on ImageNet-C; additional results cover WideResNets, ResNet-34, ViT-B/16, and CLIP ViT-B/16. We compare each base TTA method with its MIRA variant under matched settings within the pair. The original-protocol and additional learning-rate evaluations are kept distinct in Appendix D.8; absolute results from different protocols should not be compared as if their settings were identical. MIRA's probe uses only unlabeled target data. EATA's source-estimated Fisher regularizer is retained as a baseline-specific exception (Appendix D.3).

We report classification accuracy (%), mean maximum confidence (MMC), ECE with 10 equal-width bins, and NLL. MMC is interpreted jointly with accuracy, not optimized downward. For repeated runs, $\pm$ denotes sample standard deviation across seeds, with accuracy deviations in percentage points. Single-run ablations are identified as such; we do not claim statistical significance for every small difference. Implementation, hardware, and default settings are in Appendix D.

5.2 Perturbation response across architectures

Figure 4 compares accuracy and confidence across four backbones. MIRA retains the benefit of adaptation while reducing confidence–accuracy mismatch in the illustrated settings. This diagnostic does not establish a universal hyperparameter optimum; the fixed headline defaults are reported separately in Appendix D.5.

5.3 Loss design and the contribution of gating

Table 2 distinguishes the adaptation objective from its reliability weights. Direct MI minimization lowers confidence but damages accuracy and NLL; low ECE alone therefore does not imply a better predictor. The unweighted $\mathrm{HE} + \mathrm{EH}$ objective accounts for a substantial part of the ImageNet-C gain. MI–confidence gating yields a complementary accuracy/NLL trade-off, with similar or slightly worse ECE in this setting. We do not attribute the entire gain to gating. Compared with the strength-matched TENT-$2H$ baseline, the stochastic objective changes calibration as well as loss magnitude. Under separately matched gradient norms, full gating improves accuracy from 88.12 to 88.47 on CIFAR-10-C and from 68.45 to 68.87 on CIFAR-100-C (Table 3). The corresponding ECE and NLL reductions show that the incremental effect is not just a uniform change in confidence. On CIFAR-100-C,

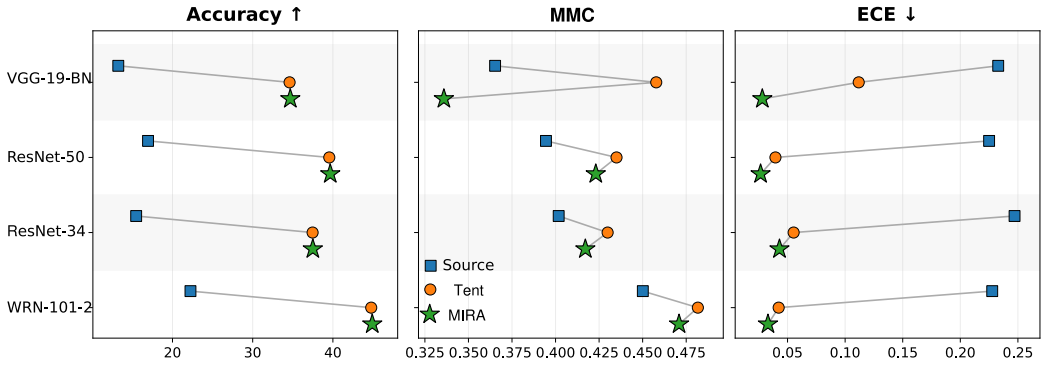


Figure 4: Cross-architecture diagnostic. From left to right the horizontal axes show accuracy (%), MMC, and ECE; rows show VGG-19-BN, ResNet-50, ResNet-34, and WRN-101-2. Markers compare Source, TENT, and MIRA.

Table 2: Single-run loss ablation on ImageNet-C with ResNet-34. MI-only minimization harms accuracy; stochastic entropy and reliability weighting have distinct roles.

Method	**Loss**	**Acc. (%)** ↑	**MMC**	**ECE** ↓	**NLL** ↓
Source	–	18.54	0.3749	0.1895	5.4921
TENT	H	34.30	0.4063	0.0642	3.4603
TENT	$2H$	36.73	0.4400	0.0732	3.2856
MIRA	H_E	34.20	0.3672	0.0421	3.4505
MIRA	E_H	34.03	0.3654	0.0419	3.4642
MIRA	MI	29.86	0.3096	0.0373	3.8238
MIRA	$H_E + E_H$	36.50	0.3975	0.0444	3.2773
MIRA	$w_{\mathrm{MI}} H_E$	34.38	0.3686	0.0422	3.4396
MIRA	$w_{\mathrm{MI}}(H_E + E_H)$	36.69	0.3986	0.0447	3.2687
MIRA	$w_{\mathrm{MI}} w_{\mathrm{conf}}(H_E + E_H)$	**36.95**	0.4004	0.0451	**3.2648**

adding the MI gate alone gives 68.62% accuracy, ECE 0.0499, and NLL 1.2058; adding confidence weighting further improves these metrics to 68.87%, 0.0441, and 1.1797. Multiplying deterministic entropy by two yields the same update as TENT under this gradient normalization, while sample-dependent weighting can change the update direction. These controls support a complementary role for gating; they do not make the gate the sole source of MIRA's improvement.

5.4 Plug-in evaluation and run-to-run variation

Table 4 reports the available multi-run comparisons. Mean accuracy, ECE, and NLL improve in the displayed pairs, but their magnitudes vary. For example, the CIFAR-100-C ROID accuracy change is only 0.05 percentage points and is smaller than the MIRA run-to-run SD; it is not decisive evidence of an accuracy gain. The additional common-learning-rate repeats show similarly small changes for some other pairs (Appendix D.8). The revised SAR integration improves ImageNet-C accuracy from 36.72 ± 0.15 to 39.28 ± 0.56, ECE from 0.0355 to 0.0290, and NLL from 3.2487 to 3.0494. The earlier indirect integration did not improve SAR accuracy on ImageNet-C and is retained as a separate comparison in Appendix D.6. This contrast makes the integration procedure important: reliability weights are estimated before SAR's optimizer perturbation and held fixed during both SAM steps. Simply nesting repeated classifier probes inside a stateful optimizer does not implement the same experiment.

Confidence need not move in one direction: CIFAR-100-C EATA's MMC increases as accuracy improves, whereas other pairs reduce MMC. ECE and NLL quantify whether confidence and the full distribution improve alongside accuracy. These are complementary outcomes, rather than a guarantee that the gate improves every metric under every protocol.

Table 3: Controlled contribution of reliability weighting after matching gradient magnitudes. These are single-run ablations on CIFAR-10-C / WRN AugMix and CIFAR-100-C / ResNeXt AugMix. Appendix B includes deterministic-entropy and MI-only-gate controls.

Dataset	Gradient-matched objective	Acc. (%) ↑	ECE ↓	NLL ↓
CIFAR-10-C	HE + EH	88.12	0.0402	0.3878
	+ MI + confidence	**88.47**	**0.0386**	**0.3786**
CIFAR-100-C	HE + EH	68.45	0.0529	1.2157
	+ MI + confidence	**68.87**	**0.0441**	**1.1797**

Table 4: Matched-pair plug-in results (mean ± sample SD over three runs). CIFAR-100-C TENT/EATA/ROID use original-protocol repeats; ImageNet-C uses the additional protocol. SAR uses separated reliability estimation. See Appendix D.8 for protocol and seed distinctions. MMC/ECE/NLL deviations retain four decimal places.

Method	Acc. (%) ↑	MMC	ECE ↓	NLL ↓
CIFAR-100-C / ResNeXt				
TENT	67.84 ± 0.05	0.7461 ± 0.0006	0.0692 ± 0.0005	1.2547 ± 0.0021
+ MIRA	$\mathbf{68.36} \pm 0.11$	0.7330 ± 0.0008	$\mathbf{0.0507} \pm 0.0008$	$\mathbf{1.2180} \pm 0.0065$
EATA	65.37 ± 0.14	0.7039 ± 0.0008	0.0559 ± 0.0006	1.3515 ± 0.0038
+ MIRA	$\mathbf{68.47} \pm 0.04$	0.7217 ± 0.0006	$\mathbf{0.0399} \pm 0.0003$	$\mathbf{1.1921} \pm 0.0025$
SAR	67.55 ± 0.11	0.7357 ± 0.0004	0.0630 ± 0.0006	1.2574 ± 0.0028
+ MIRA	$\mathbf{68.61} \pm 0.10$	0.7348 ± 0.0007	$\mathbf{0.0500} \pm 0.0006$	$\mathbf{1.2001} \pm 0.0051$
ROID	69.16 ± 0.01	0.7644 ± 0.0002	0.0735 ± 0.0003	1.1931 ± 0.0041
+ MIRA	$\mathbf{69.21} \pm 0.12$	0.7407 ± 0.0004	$\mathbf{0.0497} \pm 0.0011$	$\mathbf{1.1674} \pm 0.0049$
ImageNet-C / ResNet-50				
TENT	36.57 ± 0.26	0.4086 ± 0.0013	0.0483 ± 0.0014	3.2573 ± 0.0097
+ MIRA	$\mathbf{39.34} \pm 0.16$	0.4052 ± 0.0016	$\mathbf{0.0368} \pm 0.0020$	$\mathbf{3.0281} \pm 0.0099$
EATA	40.02 ± 0.29	0.4507 ± 0.0018	0.0549 ± 0.0020	2.9796 ± 0.0120
+ MIRA	$\mathbf{43.18} \pm 0.12$	0.4802 ± 0.0022	$\mathbf{0.0533} \pm 0.0006$	$\mathbf{2.7547} \pm 0.0005$
SAR	36.72 ± 0.15	0.3935 ± 0.0005	0.0355 ± 0.0010	3.2487 ± 0.0095
+ MIRA	$\mathbf{39.28} \pm 0.56$	0.3886 ± 0.0035	$\mathbf{0.0290} \pm 0.0021$	$\mathbf{3.0494} \pm 0.0344$
ROID	44.91 ± 0.20	0.5231 ± 0.0022	0.0758 ± 0.0007	2.6526 ± 0.0069
+ MIRA	$\mathbf{45.37} \pm 0.25$	0.4847 ± 0.0019	$\mathbf{0.0383} \pm 0.0010$	$\mathbf{2.5916} \pm 0.0066$

5.5 Transformer and vision–language generalization

With ViT-B/16 on ImageNet-C, MIRA improves accuracy, ECE, and NLL (Table 5). For CLIP, we use the released CLIPTTA architecture, data pipeline, and TENT baseline [11], with the modified one-step configuration specified in Appendix D.7. We perturb class text embeddings as the effective classifier while adapting the visual encoder's normalization parameters. This is an extension of classifier probing, rather than an exact reproduction of the full CLIPTTA method. Under the matched modified configuration, accuracy improves from 52.74 to 54.13 on CIFAR-10-C and from 27.83 to 32.71 on CIFAR-100-C, with lower ECE and NLL. On CIFAR-100-C, ECE changes from 0.5603 to 0.4784 and NLL from 8.9705 to 7.5339. The model nevertheless remains strongly miscalibrated, so this extension supports a relative improvement rather than satisfactory absolute calibration.

5.6 Stronger shifts and online adaptation protocols

Table 6 tests whether the classifier probe remains useful when target images differ substantially from ordinary photographs. MIRA improves TENT accuracy from 28.22 ± 0.42 to 29.57 ± 0.29, while reducing ECE from 0.1685 to 0.1247 and NLL from 4.1317 to 3.9175. Classifier-only stochasticity limits the scope of the reliability estimate, but does not restrict adaptation to classifier parameters: the underlying TTA method still updates its selected representation parameters. This distinction explains

Table 5: Single-run ViT/CLIP extension. CLIP uses one adaptation iteration and a $10\times$ learning-rate multiplier for both methods relative to the released TENT configuration, rather than its original 10-iteration setting.

Model / dataset	Method	Acc. (%) ↑	MMC	ECE ↓	NLL ↓
ViT / ImageNet-C	TENT	42.03	0.5178	0.1141	3.0312
	+ MIRA	**45.17**	0.4925	**0.0529**	**2.8159**
CLIP / CIFAR-10-C	TENT	52.74	0.9443	0.4170	4.6682
	+ MIRA	**54.13**	0.9387	**0.3977**	**4.5456**
CLIP / CIFAR-100-C	TENT	27.83	0.8382	0.5603	8.9705
	+ MIRA	**32.71**	0.8047	**0.4784**	**7.5339**

Table 6: ImageNet-Sketch under stronger representation shift. Mean ± sample SD over seeds 0, 1, and 2; accuracy deviations are percentage points.

Method	Acc. (%) ↑	MMC	ECE ↓	NLL ↓
Source	23.84 ± 0.35	0.4623 ± 0.0013	0.2240 ± 0.0029	5.0889 ± 0.0089
TENT	28.22 ± 0.42	0.4507 ± 0.0031	0.1685 ± 0.0014	4.1317 ± 0.0163
+ MIRA	$\mathbf{29.57 \pm 0.29}$	0.4204 ± 0.0025	$\mathbf{0.1247 \pm 0.0007}$	$\mathbf{3.9175 \pm 0.0264}$

how an inexpensive last-layer probe can guide adaptation under feature shift without capturing all feature-level uncertainty.

We also evaluate protocols with continual adaptation, reset behavior, dynamic streams, and a time budget (Appendix C) [24, 26]. Under continual CIFAR-100-C adaptation, integrating MIRA with CoTTA improves mean accuracy from 64.16 to 65.20% and reduces aggregate NLL from 1.4173 to 1.3459. Under the supplied dynamic stream, RoTTA accuracy rises from 74.89 to 75.67% and ECE falls from 0.0842 to 0.0674. The dynamic TENT pair remains near 15% accuracy despite a small improvement, showing that the gate cannot substitute for an adaptation mechanism suited to the stream. The reset-based comparison similarly improves accuracy but worsens NLL; these protocol-dependent outcomes limit a claim of uniform improvement.

The time-constrained protocol [1] accounts for adaptation speed with a constant-speed input stream. At $\eta = 1$, ImageNet-C accuracy rises from 38.97 to 39.92% and ECE falls from 0.0572 to 0.0484. ImageNet-R accuracy rises from 40.74 to 41.13% with ECE changing from 0.1330 to 0.1292. Both accuracy gains are smaller than in the corresponding unrestricted comparisons, making the cost of stochastic probing relevant to deployment.

5.7 Prediction quality and computational cost

The uncertainty-estimation comparison in Appendix E.2 includes ensembles, LLLA, ICLA, BBB, and ABNN. Those methods can improve uncertainty quality, but they do not provide the same online adaptation objective as MIRA; the comparison therefore does not isolate gating. The controlled loss and matched-pair experiments provide that evidence.

Table 7 quantifies the cost of classifier-only sampling. Relative to TENT, MIRA increases runtime from 125 to 157 seconds (25.6%) and VRAM from 2338 to 2494 MiB (6.7%), while improving accuracy, ECE, and NLL. CoTTA has slightly lower ECE in this comparison, while MIRA has better accuracy and NLL at lower cost.

Perturbation scope and sampling budget. Appendix B tests deeper-layer probing and $K = 100$–1000 sampling budgets. Perturbing the last two or three layers changes accuracy from 37.04% to 37.15% or 37.31%, but increases runtime by approximately $39\times$ or $57\times$ and worsens ECE from 0.0458 to 0.0519 or 0.0649. Thus, deeper probing captures additional variation without offering a better overall trade-off in these evaluated settings. In the separate sampling-budget experiment, increasing K from 300 to 1000 changes accuracy from 29.39 to 29.76%, with runtime increasing from 74 to 116 seconds and peak VRAM from 8,671 to 22,953 MiB. The corrected memory measurements are 5,489 MiB at $K = 100$ and 8,671 MiB at $K = 300$, using a common monitoring protocol. Larger

Table 7: Single-run accuracy–cost comparison on CIFAR-100-C/ResNeXt. Runtime and peak VRAM are measured on the same hardware setting.

Method	Acc. (%) ↑	ECE ↓	NLL ↓	Time (s)	VRAM (MiB)
TENT	67.10	0.0779	1.2873	125	2338
CoTTA	64.16	**0.0529**	1.4173	1142	3062
PETAL	62.30	0.0582	1.5059	1586	3930
$\text{MIRA}_{\text{TENT}}$	**67.75**	0.0556	**1.2401**	157	2494

K has non-monotonic predictive benefits and increasing cost; $K = 300$ is a conservative operating choice, not a uniquely optimal value.

6 Conclusion

MIRA uses cached last-layer Gaussian perturbations to guide reliable online adaptation. A stochastic entropy objective supplies the adaptation signal, while detached MI–confidence weights provide complementary sample-wise control. The evaluations support an improved accuracy–calibration trade-off across several models and protocols, with method-dependent accuracy gains and a controllable computational cost.

Limitations. Perturbation-induced MI is not calibrated posterior uncertainty or a certificate of correctness. Stable predictions can remain wrong, particularly under severe representation shift; final-layer probing cannot repair arbitrary feature collapse. The finite scale grid, thresholds α and ρ, weighting temperature τ, coefficient γ, and budget K introduce design choices. The reported construction and sampling ablations do not establish sensitivity to every threshold or grid choice. Several extension and ablation results are single-run evaluations, and small accuracy gains should not be treated as statistically conclusive. The CLIP extension and optimizer-specific integrations require their own implementation details.

Broader impact. More reliable confidence may help users identify uncertain predictions under distribution shift. Online updates can nevertheless reinforce errors or affect population groups differently; aggregate calibration on these image benchmarks does not establish fairness or safety for a downstream deployment. Monitoring, independent validation, and a conservative fallback remain necessary in consequential applications.

References

[1] Motasem Alfarra, Hani Itani, Alejandro Pardo, Shyma Yaser Alhuwaider, Merey Ramazanova, Juan Camilo Perez, Zhipeng Cai, Matthias Müller, and Bernard Ghanem. Evaluation of test-time adaptation under computational time constraints. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 976–991, 2024.

[2] Fin Amin and Jung-Eun Kim. The over-certainty phenomenon in modern test-time adaptation algorithms. *arXiv preprint arXiv:2404.16168*, 2024.

[3] Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight uncertainty in neural network. In *International conference on machine learning*, pages 1613–1622. PMLR, 2015.

[4] Dhanajit Brahma and Piyush Rai. A probabilistic framework for lifelong test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3582–3591, 2023.

[5] Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux-effortless bayesian deep learning. *Advances in neural information processing systems*, 34:20089–20103, 2021.

[6] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *international conference on machine learning*, pages 1050–1059. PMLR, 2016.

[7] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017.

[8] Dan Hendrycks, Steven Basart, Norman Mu, Saurav Kadavath, Frank Wang, Evan Dorundo, Rahul Desai, Tyler Zhu, Samyak Parajuli, Mike Guo, Dawn Song, Jacob Steinhardt, and Justin Gilmer. The many faces of robustness: A critical analysis of out-of-distribution generalization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8340–8349, 2021.

[9] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *arXiv preprint arXiv:1903.12261*, 2019.

[10] Agustinus Kristiadi, Matthias Hein, and Philipp Hennig. Being bayesian, even just a bit, fixes overconfidence in relu networks. In *International conference on machine learning*, pages 5436–5446. PMLR, 2020.

[11] Marc Lafon, Gustavo Vargas Hakim, Clément Rambour, Christian Desrosiers, and Nicolas Thome. CLIPTTA: Robust contrastive vision-language test-time adaptation. In *Advances in Neural Information Processing Systems*, volume 38, 2025.

[12] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems*, 30, 2017.

[13] Jae-Hong Lee. Bayesian weight enhancement with steady-state adaptation for test-time adaptation in dynamic environments. In *Proceedings of the 42nd International Conference on Machine Learning*, pages 32889–32906, 2025.

[14] Taeckyung Lee, Sorn Chottananurak, Taesik Gong, and Sung-Ju Lee. Aetta: Label-free accuracy estimation for test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 28643–28652, 2024.

[15] Yanghao Li, Naiyan Wang, Jianping Shi, Xiaodi Hou, and Jiaying Liu. Adaptive batch normalization for practical domain adaptation. *Pattern Recognition*, 80:109–117, 2018.

[16] Jian Liang, Ran He, and Tieniu Tan. A comprehensive survey on test-time adaptation under distribution shifts. *International Journal of Computer Vision*, 133(1):31–64, 2025.

[17] Shuaicheng Niu, Jiaxiang Wu, Yifan Zhang, Yaofo Chen, Shijian Zheng, Peilin Zhao, and Mingkui Tan. Efficient test-time model adaptation without forgetting. In *International conference on machine learning*, pages 16888–16905. PMLR, 2022.

[18] Shuaicheng Niu, Jiaxiang Wu, Yifan Zhang, Zhiquan Wen, Yaofo Chen, Peilin Zhao, and Mingkui Tan. Towards stable test-time adaptation in dynamic wild world. *arXiv preprint arXiv:2302.12400*, 2023.

[19] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, David Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. *Advances in neural information processing systems*, 32, 2019.

[20] Hidetoshi Shimodaira. Improving predictive inference under covariate shift by weighting the log-likelihood function. *Journal of statistical planning and inference*, 90(2):227–244, 2000.

[21] Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei Efros, and Moritz Hardt. Test-time training with self-supervision for generalization under distribution shifts. In *International conference on machine learning*, pages 9229–9248. PMLR, 2020.

[22] Mingkui Tan, Guohao Chen, Jiaxiang Wu, Yifan Zhang, Yaofo Chen, Peilin Zhao, and Shuaicheng Niu. Uncertainty-calibrated test-time model adaptation without forgetting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

[23] Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. *arXiv preprint arXiv:2006.10726*, 2020.

[24] Qin Wang, Olga Fink, Luc Van Gool, and Dengxin Dai. Continual test-time domain adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7201–7211, 2022.

[25] Zixin Wang, Yadan Luo, Liang Zheng, Zhuoxiao Chen, Sen Wang, and Zi Huang. In search of lost online test-time adaptation: A survey. *International Journal of Computer Vision*, 133(3):1106–1139, 2025.

[26] Longhui Yuan, Binhui Xie, and Shuang Li. Robust test-time adaptation in dynamic scenarios. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15922–15932, 2023.

[27] Marvin Zhang, Sergey Levine, and Chelsea Finn. Memo: Test time robustness via adaptation and augmentation. *Advances in neural information processing systems*, 35:38629–38642, 2022.

[28] Qingyang Zhang, Yatao Bian, Xinke Kong, Peilin Zhao, and Changqing Zhang. Come: Test-time adaption by conservatively minimizing entropy. *arXiv preprint arXiv:2410.10894*, 2024.

[29] Maksim Zhdanov, Stanislav Dereka, and Sergey Kolesnikov. Identity curvature laplace approximation for improved out-of-distribution detection. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 7019–7028. IEEE, 2025.

[30] Aurick Zhou and Sergey Levine. Bayesian adaptation for covariate shift. *Advances in neural information processing systems*, 34:914–927, 2021.

[31] Lihua Zhou, Mao Ye, Shuaifeng Li, Nianxin Li, Xiatian Zhu, Lei Deng, Hongbin Liu, and Zhen Lei. Bayesian test-time adaptation for vision-language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 29999–30009, 2025.

A Rationale and local perturbation analysis

The analysis below is conditional on fixed cached features and fixed scale factors for a target batch. Gaussian classifier noise is independent and centered. It describes local sensitivity, not correctness or a fitted posterior.

A.1 Agreement and pairwise margins

Let $\hat{y}_i = \arg\max_c z_{i,c}$, and let $m_{ij} = z_{i,\hat{y}_i} - z_{i,j} \geq 0$ be its margin against class j. Under Eq. (2), independence of the noise in the two classifier rows gives

$$\Delta m_{ij} \sim \mathcal{N}(0, \sigma^2 s_{ij}^2), \qquad s_{ij}^2 = \sum_d \big(A_{\mathcal{B}}[\hat{y}_i, d]^2 + A_{\mathcal{B}}[j, d]^2\big) h_{i,d}^2 + a_{\mathcal{B}}[\hat{y}_i]^2 + a_{\mathcal{B}}[j]^2. \quad (9)$$

For $s_{ij} > 0$, standardizing the Gaussian variable yields

$$\Pr(m_{ij} + \Delta m_{ij} < 0) = \Phi(-m_{ij}/(\sigma s_{ij}))\,. \quad (10)$$

This is a pairwise boundary-crossing probability, not the full multiclass error probability. A union bound can upper-bound changes from the unperturbed top-1 class by the sum over competitors. The implemented agreement instead compares each sampled top-1 class with the Monte Carlo mean top-1 class. Its exact empirical interpretation is the fraction of samples retaining that reference class; $1 - A(\sigma)$ is the batch-average disagreement rate. These reference classes need not coincide with the correct label or with the unperturbed prediction.

A.2 MI as local distributional sensitivity

Let $p_i = \mathrm{softmax}(z_i)$, and let δz_i be a small, centered logit perturbation with covariance $\Sigma_{z,i}$. For finite logits and positive probabilities, the softmax Jacobian is $J_i = \mathrm{diag}(p_i) - p_i p_i^\top$. To first order, $\delta p_i = J_i \delta z_i$. The Hessian of entropy at p_i is $-\mathrm{diag}(1/p_i)$. Expanding both $H(\mathbb{E}[p_i + \delta p_i])$ and $\mathbb{E}[H(p_i + \delta p_i)]$ to second order cancels their mean terms and gives

$$\mathrm{MI}_{i,\sigma} \approx \frac{1}{2} \mathrm{tr}\big(\mathrm{diag}(1/p_i) J_i \Sigma_{z,i} J_i^\top\big) = \frac{1}{2} \mathrm{tr}(J_i \Sigma_{z,i})\,. \quad (11)$$

The equality uses $J_i \mathrm{diag}(1/p_i) J_i = J_i$ and cyclic invariance of the trace. Thus MI is locally sensitive to the perturbation covariance and the predictive geometry. The approximation requires small perturbations and does not characterize the large-radius collapse regime. Monte Carlo estimates also have finite-sampling variability.

A.3 Why exponential reliability weights?

For fixed scores m_i, consider weights $q_i \geq 0$ on the probability simplex and the entropy-regularized objective

$$\min_q \sum_i q_i m_i + \tau \sum_i q_i \log(B q_i), \qquad \sum_i q_i = 1, \quad \tau > 0. \quad (12)$$

The stationary condition with Lagrange multiplier λ is $m_i + \tau(\log(Bq_i) + 1) + \lambda = 0$. Strict convexity gives the unique solution $q_i = e^{-m_i/\tau} / \sum_j e^{-m_j/\tau}$, and $w_i^{\mathrm{MI}} = B q_i$ has unit batch mean. Subtracting the common batch mean from the scores leaves these weights unchanged. Multiplying by confidence and renormalizing supplies the additional confidence gate; both factors are detached for adaptation. This variational rationale does not prove that stability implies correctness or that weighted gradients have the same norm as unweighted gradients. The entropy-matched diagnostic in Appendix E and the controlled ablations in Appendix B provide empirical support for the reliability interpretation.

B Additional ablation studies

These ablations are single-run comparisons unless an uncertainty estimate is explicitly shown. They test the objective, perturbation construction, scope, and Monte Carlo budget; they do not establish sensitivity to every hyperparameter.

B.1 Gradient-matched reliability gating

The compared objectives use separately matched gradient magnitudes. Multiplying deterministic entropy by two then gives the same normalized update as TENT, whereas the stochastic objective and sample-wise weights can change the update direction. Unit-mean sample weights alone do not ensure matched gradient norms.

Table 8: Gradient-matched loss ablation on CIFAR-10-C / WRN AugMix.

Objective	Acc. (%) ↑	ECE ↓	NLL ↓
TENT / TENT-2H	87.75	0.0635	0.4444
HE + EH	88.12	0.0402	0.3878
+ MI gate	88.38	0.0393	0.3815
+ MI + confidence	88.47	0.0386	0.3786

Table 9: Gradient-matched loss ablation on CIFAR-100-C / ResNeXt AugMix.

Objective	Acc. (%) ↑	ECE ↓	NLL ↓
TENT	67.90	0.0692	1.2523
HE + EH	68.45	0.0529	1.2157
+ MI gate	68.62	0.0499	1.2058
+ MI + confidence	68.87	0.0441	1.1797

The stochastic objective provides the main gain and the gates give complementary improvements after matching gradient magnitudes. Thus, their effects are not explained solely by a stronger global entropy coefficient.

B.2 Perturbation design sensitivity

The base-scale rule and its scalar/class-wise/feature-wise parameterization are two separate construction axes. Each is varied while retaining the other settings on CIFAR-100-C, severity 5. Accuracy is reported in percent, runtime in seconds, and memory in MiB.

Table 10: Scale-parameterization ablation on CIFAR-100-C.

Construction	Acc. (%)	MMC	ECE	NLL	Time (s)	VRAM (MiB)
TENT	63.72	0.7184	0.0816	1.4061	78	931
Feature-wise	64.42	0.6786	0.0383	1.3383	100	1175
Scalar	64.42	0.6585	0.0288	1.3361	100	1175
Class-wise	64.42	0.6585	0.0287	1.3356	103	1047

Table 11: Base-scale ablation on CIFAR-100-C.

Construction	Acc. (%)	MMC	ECE	NLL	Time (s)	VRAM (MiB)
Source	53.25	0.7095	0.1770	2.1697	43	555
TENT	63.71	0.7183	0.0816	1.4059	76	803
He-in	64.42	0.6786	0.0382	1.3382	100	1135
Fan-in	64.37	0.6598	0.0200	1.3287	103	1047
Weight magnitude	64.36	0.6559	0.0236	1.3409	106	1047
Absolute	64.32	0.6439	0.0154	1.3291	100	1135

All tested constructions give similar accuracy, but calibration varies more noticeably. The headline configuration uses feature-wise He-in scaling; we do not select the best ECE rule separately for each test scenario.

B.3 Perturbation scope

We compare the final classifier with the last two or three layers on ImageNet-C.

Table 12: Predictive metrics for different perturbation scopes on ImageNet-C.

Scope	Acc. (%)	MMC	ECE	NLL
Classifier only	37.04	0.4013	0.0458	3.2690
Last 2 layers	37.15	0.3576	0.0519	3.2372
Last 3 layers	37.31	0.3592	0.0649	3.2332

Table 13: Computational cost of the same scope comparisons.

Scope	Time (s)	VRAM (MiB)	Perturbed parameters
Classifier only	379.9	8180.4	513,000
Last 2 layers	14,818.9	8212.3	2,872,296
Last 3 layers	21,572.5	8212.3	5,231,592

Deeper probes give small accuracy/NLL improvements but worsen ECE and increase runtime by approximately $39\times$ and $57\times$. Classifier-only probing offers the more favorable trade-off among these evaluated settings, without capturing all representation-level uncertainty.

B.4 Monte Carlo sampling budget

Table 14: Monte Carlo budget on ImageNet-C. Corrected peak-memory measurements use the common monitoring protocol.

K	Acc. (%)	ECE	NLL	Time (s)	Peak VRAM (MiB)
100	29.55	0.0410	3.6454	60	5,489
200	29.55	0.0408	3.6407	66	7,175
300	29.39	0.0420	3.6374	74	8,671
500	29.29	0.0443	3.6511	88	11,939
1000	29.76	0.0381	3.6387	116	22,953

Runtime and memory increase with K, while predictive metrics are non-monotonic and their changes are small. $K = 300$ is a conservative operating choice rather than a uniquely optimal budget.

C Additional experiments

We retain the available evaluations under continual adaptation, dynamic target streams, stronger representation shift, and computational time constraints. Results are single-run values except for the explicitly repeated ImageNet-Sketch comparison.

C.1 Continual test-time adaptation

The continual CIFAR-100-C experiment uses severity 5. Table 15 reports per-corruption accuracy (%) without compressing all corruption names into one wide row.

Table 15: Continual CIFAR-100-C / ResNeXt results.

Corruption	PETAL	$\text{MIRA}_{\text{TENT}}$	CoTTA	CoTTA + MIRA
Gaussian noise	59.50	62.96	58.79	56.83
Shot noise	62.62	65.55	60.93	59.69
Impulse noise	60.64	67.75	58.57	57.43
Defocus blur	72.18	74.76	71.39	72.44
Glass blur	62.51	63.30	58.70	58.72
Motion blur	71.28	72.74	69.45	70.94
Zoom blur	72.62	74.55	71.08	72.79
Snow	66.38	69.88	63.29	65.61
Frost	67.50	69.29	64.34	66.06
Fog	58.58	67.20	55.12	59.34
Brightness	73.03	75.88	72.65	74.09
Contrast	70.89	72.52	67.13	70.29
Elastic transform	66.59	67.37	62.67	65.29
Pixelation	70.68	71.36	67.47	68.27
JPEG compression	65.95	63.98	60.81	60.27
Average accuracy	66.73	69.27	64.16	65.20
Average ECE	0.0591	0.0199	0.0528	0.0342

The integrated CoTTA pair improves average accuracy and calibration. Its reported aggregate MMC changes from 0.6170 to 0.6787 and NLL from 1.4173 to 1.3459. The aggregate comparison reports ECE 0.0529, while the per-corruption summary reports 0.0528. This small reporting discrepancy should be resolved from the original metric outputs before comparing the two summaries.

C.2 Alternative reset-based protocol

Table 16: Reset-based PTTA comparison on ImageNet-C / ResNet-50.

Method	Acc. (%)	MMC	ECE	NLL
PTTA	37.50	0.3750	0.0284	2.7080
+ MIRA	40.67	0.4067	0.0278	2.9286

This comparison improves accuracy and slightly lowers ECE, while NLL worsens. It is an example of a trade-off, rather than evidence of improvement in every probabilistic metric.

C.3 Stronger representation shift: ImageNet-Sketch

The three-run results over seeds 0, 1, and 2 are reported in Table 6 in the main text. MIRA improves TENT mean accuracy by 1.35 percentage points while reducing ECE and NLL. This indicates that the probe can remain informative under substantial representation shift; it does not remove the limitation of final-layer probing.

C.4 Dynamic target streams

The practical TTA (PTTA) setting considers temporal distribution changes and correlations [26]. The supplied dynamic-stream evaluation compares TENT and RoTTA with their MIRA variants.

Table 17: Dynamic-stream evaluation.

Method	Acc. (%)	MMC	ECE	NLL
TENT	14.75	0.9675	0.8201	17.6335
+ MIRA	15.16	0.9650	0.8134	17.0462
RoTTA	74.89	0.8331	0.0842	0.8933
+ MIRA	75.67	0.8241	0.0674	0.8622

MIRA improves the reported metrics for both underlying methods, but the TENT pair still has low accuracy and extreme overconfidence. The gate does not by itself resolve unstable adaptation in this stream.

C.5 Computationally realistic online evaluation

The time-constrained protocol of Alfarra et al. [1] accounts for adaptation speed by limiting updates in a constant-speed input stream. The reported realistic evaluation uses $\eta = 1$; the conventional comparison does not impose this time constraint. ImageNet-R evaluates naturally occurring renditions [8].

Table 18: Conventional and time-constrained ($\eta = 1$) online evaluation.

Dataset	Method	Realistic	Acc. (%)	MMC	ECE	NLL
ImageNet-C	TENT	No	42.71	0.5203	0.0935	2.9071
	+ MIRA	No	45.02	0.5007	0.0573	2.7948
	TENT	Yes	38.97	0.4444	0.0572	3.0884
	+ MIRA	Yes	39.92	0.4412	0.0484	3.0072
ImageNet-R	TENT	No	41.84	0.5834	0.1649	3.2015
	+ MIRA	No	44.10	0.5771	0.1361	2.9875
	TENT	Yes	40.74	0.5404	0.1330	3.2062
	+ MIRA	Yes	41.13	0.5405	0.1292	3.1631

The gains become smaller under the time constraint, while their directions remain consistent for the reported accuracy, ECE, and NLL values.

C.6 Accuracy–cost trade-off

Table 7 contains the same matched-hardware runtime and VRAM comparison. For completeness, the corresponding MMC values are TENT 0.7474, CoTTA 0.6170, PETAL 0.5816, and MIRA$_{\mathrm{TENT}}$ 0.7310. These single-run cost results do not supply error bars for the separate loss or implementation ablations.

D Experimental and Implementation Details

D.1 Experimental Setup

We evaluate MIRA under the source-free online test-time adaptation setting. For fair comparison, all methods use the same backbone architecture, source training setup, preprocessing, and target corruption streams whenever applicable. Unless otherwise stated, all experiments on CIFAR-10-C, CIFAR-100-C, and ImageNet-C use corruption severity 5, and all method-specific hyperparameters are fixed across corruption types.

All experiments are conducted on NVIDIA RTX A6000 GPUs. Runtime and peak-memory measurements are reported for the available cost comparisons; these do not constitute a complete accounting of total project compute.

D.2 Evaluation Metrics

We report classification accuracy, maximum mean confidence (MMC), expected calibration error (ECE), negative log-likelihood (NLL), and, where applicable, selective-prediction metrics.

Let p_i denote the predictive probability for sample i, $\hat{y}_i = \arg\max_c p_{i,c}$, and $c_i = \max_c p_{i,c}$. Accuracy is the fraction of correct top-1 predictions, while $\mathrm{MMC} = \frac{1}{N}\sum_i c_i$ measures average predictive confidence.

ECE is computed using 10 equal-width confidence bins following [7]:

$$\mathrm{ECE} = \sum_{m=1}^{10} \frac{|\mathcal{B}_m|}{N} \left|\mathrm{acc}(\mathcal{B}_m) - \mathrm{conf}(\mathcal{B}_m)\right|.$$

NLL is computed as

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log p_{i,y_i},$$

with probabilities lower-bounded by 10^{-12} for numerical stability.

D.3 Baseline Implementation Details

We compare MIRA with TENT, EATA, SAR, and ROID under the standard online TTA protocol. We follow the optimizer settings of the corresponding base methods. For the CIFAR-100-C experiments, TENT uses SGD with a learning rate of 1×10^{-3}, while EATA, SAR, and ROID use a learning rate of 2.5×10^{-4}. We use momentum 0.9 and zero weight decay unless otherwise specified. When MIRA is integrated with a base TTA method, we retain the optimizer and learning rate of that method rather than retuning them.

For EATA, we additionally use Fisher regularization of 1.0 and confidence margin $d = 0.4$, with the Fisher information matrix estimated from 2000 source samples. This is a source-dependent component of the EATA baseline; MIRA adds no source-data requirement.

We also compare against uncertainty-estimation baselines including Deep Ensemble (DE), LLLA, ICLA, BBB, and ABNN. DE averages predictions from independently trained source models. LLLA follows the official Laplace implementation, while the remaining methods follow their released implementations or published settings using the same backbone and evaluation protocol whenever possible.

D.4 Perturbation Scale Construction

Let $W \in \mathbb{R}^{C\times D}$ denote the final classifier weight and $h_i \in \mathbb{R}^D$ the cached feature of target sample i. All headline experiments use a He-style base scale with a feature-wise, batch-dependent factor:

$$s_0 = \sqrt{\frac{2}{D}}, \qquad r_{\mathcal{B},d} = \frac{1}{B}\sum_{i=1}^{B} |h_{i,d}|,$$

$$g_{\mathcal{B},d} = \mathrm{clip}\left(\frac{r_{\mathcal{B},d}}{D^{-1}\sum_{j=1}^{D} r_{\mathcal{B},j}}, 0.25,\ 4.0 \right).$$

The perturbation factors are

$$A_{\mathcal{B}}[c,d] = s_0 g_{\mathcal{B},d}, \qquad a_{\mathcal{B}}[c] = s_0,$$

where $g_{\mathcal{B},d}$ is shared across output classes. The He-style factor accounts for classifier input dimension, while the feature-wise factor adapts the perturbation geometry to the current target representation. Neither factor is tuned per dataset or model; only the global perturbation radius σ is selected online.

D.5 Hyperparameter Settings

We use the same default MIRA configuration across datasets, architectures, corruption types, and base TTA methods.

Table 19: Default hyperparameters used in the main experiments.

Parameter	**Default**	**Role**
α	0.03	Non-trivial MI threshold
ρ	0.90	Prediction-agreement threshold
τ	0.25	MI-weighting temperature
γ	1.0	Coefficient of expected entropy
K	300	Monte Carlo samples
Batch size	64	Target adaptation batch size
Feature clipping	$[0.25, 4.0]$	Bounds on $g_{\mathcal{B},d}$
Noise across scales	Shared	Same MC samples across candidate scales
σ interpolation	Enabled	Interpolation between candidate scales

For all headline experiments, we use the fixed candidate scale set

$$\mathcal{G} = \{10^{-5}, 10^{-4}, 3{\times}10^{-4}, 10^{-3}, 3{\times}10^{-3}, 10^{-2}, 3{\times}10^{-2}, 5{\times}10^{-2}, \\ 10^{-1}, 3{\times}10^{-1}, 5{\times}10^{-1}, 1, 2, 3, 4, 5\}.$$

The base scale rules and the scalar/class-wise/feature-wise parameterizations are separate construction axes, as evaluated in Appendix B.

The same Monte Carlo noise samples are reused across candidate scales within each target batch. Since the backbone features are cached, scale evaluation requires only repeated final-layer operations.

The normalized MI threshold $\alpha = 0.03$ detects the earliest non-trivial stochastic response, while $\rho = 0.90$ restricts the empirical top-1 disagreement rate to approximately 10%. We use $\tau = 0.25$ for MI-based reweighting and $K = 300$ Monte Carlo samples. No corruption-specific or dataset-specific tuning is performed for these headline defaults. The cross-architecture plot is a separate diagnostic, rather than a threshold-sensitivity study.

Practical interpretation. α is dimensionless because MI is normalized by its range in the current batch. ρ bounds disagreement relative to the Monte Carlo mean, and should be interpreted as a class-preserving constraint. The defaults seek an early detectable response, rather than maximal disagreement. The author response recommends relaxing α before lowering ρ when the constraints conflict; this is practitioner guidance, not an additional evaluated tuning procedure. The reported implementation enables interpolation between grid candidates.

D.6 SAR integration

The revised integration estimates sample reliability at the unperturbed state, detaches the weights, and restores the classifier before SAR's sharpness-aware minimization (SAM) update. The weights are fixed and applied to the sample-wise entropy loss in both SAM steps. This preserves SAR's

original two-step optimization while separating it from repeated classifier probing. The revised three-run comparison gives CIFAR-100-C accuracy 67.55 ± 0.11 versus 68.61 ± 0.10, and ImageNet-C accuracy 36.72 ± 0.15 versus 39.28 ± 0.56 (base versus MIRA). The associated MMC/ECE/NLL deviations are retained in Table 4. The earlier indirect integration is reported separately in Table 20; it should not be conflated with this revised update.

D.7 CLIP implementation and configuration

We use CLIP ViT-B/16 with the released CLIPTTA architecture, data pipeline, and TENT implementation [11]. Logits are based on similarities between normalized visual features and class-specific text embeddings. The text-embedding matrix is treated as the effective classifier and is perturbed only for reliability probing; its unperturbed embeddings remain fixed. Adaptation updates the visual encoder's normalization parameters.

A sanity check of the released configuration reproduced TENT accuracies of approximately 56.3% and 31.5% on CIFAR-10-C and CIFAR-100-C. The final extension instead uses one adaptation iteration per batch (the released setting uses ten) and increases the released TENT learning rate by $10\times$ for both TENT and MIRA. Table 5 reports this matched modified configuration, with baseline accuracies 52.74% and 27.83%. These values should not be mixed with the released-configuration sanity check. The extension probes an effective similarity classifier and is not a one-to-one implementation of a learned linear head or a reproduction of full CLIPTTA.

D.8 Repeat protocols and uncertainty reporting

For $n = 3$ independent runs, the reported sample SD is $s = \sqrt{\sum_{r=1}^{3}(v_r - \bar{v})^2/(3-1)}$. This is distinct from standard error $s/\sqrt{3}$. Every uncertainty estimate reported here is SD, rather than standard error or a confidence interval. Accuracy means and SD are in percent and percentage points, respectively; a fractional accuracy SD must be multiplied by 100 when its mean is converted to percent. All MMC, ECE, and NLL deviations retain their native units.

Original-protocol CIFAR-100-C repeats. TENT uses learning rate 10^{-3}; EATA/SAR/ROID use 2.5×10^{-4} in the stated original settings. The original-protocol CIFAR-100-C repeats use seeds 1, 2, and 3 for TENT, EATA, SAR, and ROID. Table 4 uses these original-protocol TENT/EATA/ROID aggregates and separately uses the revised SAR integration. The older original-protocol SAR result is 65.3633 ± 0.1498 versus 66.7767 ± 0.0833 in accuracy, with ECE 0.0567 ± 0.0005 versus 0.0339 ± 0.0002 and NLL 1.3523 ± 0.0032 versus 1.2647 ± 0.0021. Its MMC is 0.7049 ± 0.0006 versus 0.6973 ± 0.0004.

Additional common-learning-rate repeats. The earlier additional evaluation uses learning rate 10^{-3} and seeds 2, 4, and 6, with matched learning rate, batch size, and other settings within every baseline/MIRA pair. It is a separate protocol, rather than a rerun of the original lower-learning-rate CIFAR EATA/SAR/ROID settings.

Table 20: Additional CIFAR-100-C / ResNeXt repeats at learning rate 10^{-3} (mean $\pm$ sample SD). SAR here uses the earlier indirect integration.

Method	Acc. (%)	MMC	ECE	NLL
TENT	67.87 ± 0.05	0.7461 ± 0.0006	0.0689 ± 0.0008	1.2554 ± 0.0004
+ MIRA	68.68 ± 0.04	0.7310 ± 0.0008	0.0457 ± 0.0005	1.2000 ± 0.0018
EATA	67.55 ± 0.12	0.7329 ± 0.0007	0.0606 ± 0.0007	1.2559 ± 0.0032
+ MIRA	67.69 ± 0.15	0.7298 ± 0.0002	0.0541 ± 0.0015	1.2339 ± 0.0058
SAR (indirect)	67.48 ± 0.08	0.7351 ± 0.0006	0.0630 ± 0.0009	1.2604 ± 0.0001
+ MIRA (indirect)	67.58 ± 0.11	0.7101 ± 0.0005	0.0383 ± 0.0010	1.2340 ± 0.0018
ROID	62.33 ± 0.28	0.7466 ± 0.0007	0.1236 ± 0.0021	1.5309 ± 0.0149
+ MIRA	62.50 ± 0.15	0.7229 ± 0.0004	0.0982 ± 0.0011	1.4784 ± 0.0055

The corresponding ImageNet-C additional-protocol TENT/EATA/ROID aggregates are in the main table. Before separated SAR integration, ImageNet-C SAR versus MIRA accuracy was 36.72 ± 0.1453

versus 36.54 ± 0.2639, MMC 0.3935 ± 0.0005 versus 0.3525 ± 0.0019, ECE 0.0355 ± 0.0010 versus 0.0314 ± 0.0010, and NLL 3.2487 ± 0.0095 versus 3.2638 ± 0.0177. This early integration improved ECE but not accuracy or NLL, motivating the revised SAR update described in Appendix D.6. The construction, loss, scope, sampling-budget, ViT/CLIP, and protocol-extension tables report single-run results. We do not infer their statistical significance from these separate repeats.

Table 21: Additional protocol: CIFAR-100-C (seeds 2,4,6). Rounded per-seed outputs; aggregates use full-precision source outputs.

Method	Seed	Acc. (%)	MMC	ECE	NLL
TENT	2	67.83	0.7466	0.0697	1.2557
TENT	4	67.87	0.7464	0.0689	1.2549
TENT	6	67.92	0.7454	0.0680	1.2555
TENT + MIRA	2	68.65	0.7315	0.0458	1.2009
TENT + MIRA	4	68.66	0.7313	0.0461	1.1979
TENT + MIRA	6	68.73	0.7301	0.0451	1.2012
EATA	2	67.58	0.7333	0.0608	1.2560
EATA	4	67.42	0.7321	0.0612	1.2590
EATA	6	67.65	0.7334	0.0599	1.2526
EATA + MIRA	2	67.51	0.7300	0.0558	1.2403
EATA + MIRA	4	67.75	0.7298	0.0535	1.2291
EATA + MIRA	6	67.80	0.7296	0.0529	1.2324
SAR	2	67.43	0.7353	0.0634	1.2604
SAR	4	67.44	0.7356	0.0636	1.2603
SAR	6	67.57	0.7344	0.0619	1.2604
SAR + MIRA	2	67.56	0.7106	0.0386	1.2339
SAR + MIRA	4	67.49	0.7100	0.0392	1.2358
SAR + MIRA	6	67.70	0.7097	0.0372	1.2322
ROID	2	62.28	0.7467	0.1243	1.5337
ROID	4	62.63	0.7473	0.1213	1.5148
ROID	6	62.08	0.7459	0.1253	1.5442
ROID + MIRA	2	62.49	0.7230	0.0985	1.4767
ROID + MIRA	4	62.65	0.7232	0.0970	1.4739
ROID + MIRA	6	62.36	0.7224	0.0991	1.4845

Table 22: Additional protocol: ImageNet-C (seeds 2,4,6). Rounded per-seed outputs; aggregates use full-precision source outputs.

Method	Seed	Acc. (%)	MMC	ECE	NLL
TENT	2	36.49	0.4073	0.0496	3.2635
TENT	4	36.86	0.4098	0.0469	3.2462
TENT	6	36.36	0.4087	0.0485	3.2623
TENT + MIRA	2	39.29	0.4037	0.0381	3.0207
TENT + MIRA	4	39.52	0.4068	0.0377	3.0394
TENT + MIRA	6	39.22	0.4050	0.0345	3.0243
EATA	2	39.82	0.4488	0.0567	2.9920
EATA	4	40.36	0.4523	0.0528	2.9680
EATA	6	39.89	0.4511	0.0551	2.9787
EATA + MIRA	2	43.05	0.4778	0.0530	2.7552
EATA + MIRA	4	43.28	0.4806	0.0529	2.7546
EATA + MIRA	6	43.22	0.4822	0.0539	2.7542
SAR	2	36.71	0.3931	0.0366	3.2388
SAR	4	36.87	0.3940	0.0345	3.2497
SAR	6	36.58	0.3935	0.0355	3.2577
SAR + MIRA	2	36.41	0.3508	0.0323	3.2715
SAR + MIRA	4	36.84	0.3546	0.0316	3.2435
SAR + MIRA	6	36.36	0.3520	0.0304	3.2764
ROID	2	44.82	0.5208	0.0754	2.6482
ROID	4	45.14	0.5252	0.0754	2.6491
ROID	6	44.76	0.5233	0.0766	2.6605
ROID + MIRA	2	45.23	0.4827	0.0393	2.5881
ROID + MIRA	4	45.66	0.4865	0.0373	2.5875
ROID + MIRA	6	45.23	0.4849	0.0382	2.5993

Table 23: Original protocol: CIFAR-100-C (seeds 1,2,3). Rounded per-seed outputs; aggregates use full-precision source outputs.

Method	Seed	Acc. (%)	MMC	ECE	NLL
TENT	1	67.90	0.7464	0.0692	1.2523
TENT	2	67.83	0.7466	0.0697	1.2557
TENT	3	67.80	0.7454	0.0688	1.2562
TENT + MIRA	1	68.48	0.7331	0.0504	1.2126
TENT + MIRA	2	68.31	0.7321	0.0501	1.2162
TENT + MIRA	3	68.28	0.7337	0.0517	1.2252
EATA	1	65.52	0.7041	0.0556	1.3493
EATA	2	65.35	0.7046	0.0565	1.3493
EATA	3	65.25	0.7031	0.0555	1.3558
EATA + MIRA	1	68.51	0.7221	0.0402	1.1893
EATA + MIRA	2	68.46	0.7221	0.0400	1.1940
EATA + MIRA	3	68.43	0.7210	0.0396	1.1930
SAR	1	65.53	0.7052	0.0564	1.3494
SAR	2	65.32	0.7053	0.0572	1.3518
SAR	3	65.24	0.7042	0.0564	1.3557
SAR + MIRA	1	66.87	0.6976	0.0340	1.2623
SAR + MIRA	2	66.75	0.6968	0.0336	1.2662
SAR + MIRA	3	66.71	0.6974	0.0340	1.2657
ROID	1	69.17	0.7641	0.0731	1.1951
ROID	2	69.16	0.7645	0.0736	1.1884
ROID	3	69.15	0.7645	0.0737	1.1959
ROID + MIRA	1	69.32	0.7403	0.0485	1.1693
ROID + MIRA	2	69.21	0.7411	0.0499	1.1619
ROID + MIRA	3	69.09	0.7406	0.0506	1.1711

E Uncertainty Analysis

E.1 Epistemic and Aleatoric Uncertainty

Although MIRA does not explicitly fit a Bayesian posterior, its stochastic last-layer perturbations provide a useful operational decomposition of predictive uncertainty. Given the Monte Carlo predictions, the predictive entropy

$$H(\mathbb{E}[p])$$

captures the total uncertainty of the averaged prediction, while the expected entropy

$$\mathbb{E}[H(p)]$$

measures the average uncertainty of individual perturbed classifiers. Their difference,

$$\mathrm{MI} = H(\mathbb{E}[p]) - \mathbb{E}[H(p)],$$

quantifies disagreement across the perturbed classifiers. Accordingly, we interpret $\mathbb{E}[H(p)]$ as an *aleatoric-like* component and MI as an *epistemic-like* component reflecting local prediction instability under classifier perturbations.

Figure 5 visualizes these quantities across CIFAR-10-C, CIFAR-100-C, and ImageNet-C. All three uncertainty measures are related to prediction difficulty, but predictive entropy alone does not fully characterize reliability. In particular, MI captures variation across perturbed classifiers that is not directly observable from the entropy of the mean prediction.

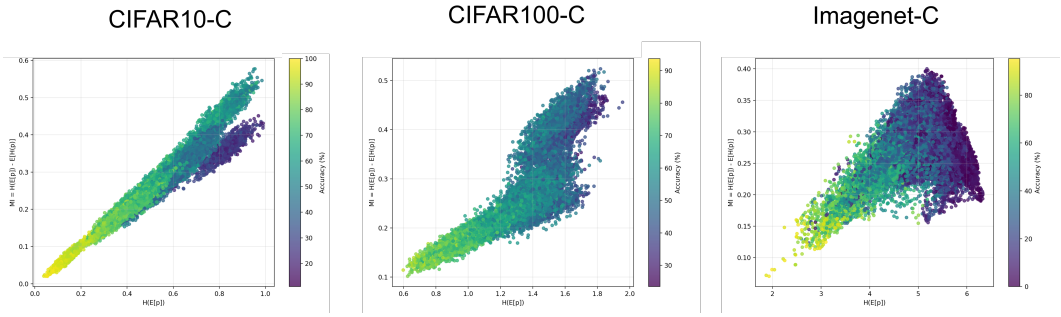


Figure 5: Visualization of predictive entropy $H(\mathbb{E}[p])$, expected entropy $\mathbb{E}[H(p)]$, and their difference as mutual information. Expected entropy and MI are interpreted as aleatoric-like and epistemic-like uncertainty components, respectively, under the local stochastic perturbation model.

To examine whether MI provides information beyond predictive entropy, we compare batches with similar $H(\mathbb{E}[p])$. As shown in Figure 6, batches with larger MI tend to exhibit lower accuracy than entropy-matched batches with smaller MI. Thus, even when the overall predictive entropy is comparable, perturbation-induced disagreement can distinguish relatively stable predictions from locally brittle ones.

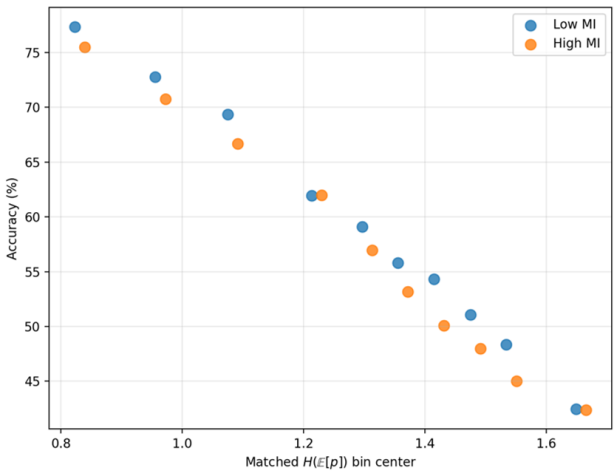


Figure 6: Relationship between batch accuracy and mutual information under matched entropy levels over all corruption types combined. Even when batches are matched by the same $H(\mathbb{E}[p])$ bin center, accuracy differs systematically with mutual information: low-MI batches tend to achieve higher accuracy than high-MI batches. This suggests that mutual information provides additional reliability information beyond predictive entropy alone.

This distinction is directly relevant to test-time adaptation. A prediction with low or moderate entropy may still be unreliable if it exhibits high MI, indicating sensitivity to small classifier perturbations. Conversely, low-MI predictions remain more consistent within the local perturbation neighborhood. These observations support the use of MI as a complementary reliability signal for MIRA, allowing unstable samples to be down-weighted before they drive entropy-based adaptation.

E.2 Comparison with uncertainty-estimation methods

The available single-run comparison evaluates DE, LLLA, ICLA, BBB, and ABNN against MIRA. MIRA adapts online, whereas these baselines estimate uncertainty under different inference procedures; this comparison is not a controlled isolation of the reliability gate. Its source-model values differ from the plug-in experiment, so the two evaluations should not be pooled.

Table 24: Uncertainty-estimation comparison on CIFAR-100-C / WRN-40-2.

Method	Acc. (%)	MMC	ECE	NLL
MAP	53.89	0.7041	0.1653	2.0589
Deep ensemble	58.11	0.6269	0.0504	1.6998
LLLA	53.04	0.6282	0.0978	1.9300
ICLA	53.89	0.4449	0.0956	1.8801
BBB	34.45	0.5445	0.2001	3.7816
ABNN	51.79	0.6568	0.1389	2.2254
$\text{MIRA}_{\text{TENT}}$	63.91	0.6701	0.0390	1.3505

Table 25: Uncertainty-estimation comparison on ImageNet-C / ResNet-50.

Method	Acc. (%)	MMC	ECE	NLL
MAP	20.84	0.2246	0.0584	4.6490
Deep ensemble	25.32	0.1874	0.0750	4.1725
LLLA	21.40	0.1948	0.0431	4.4985
ICLA	21.43	0.1879	0.0468	4.5034
BBB	20.57	0.3079	0.1035	4.8407
ABNN	23.26	0.2968	0.0634	4.2755
$\text{MIRA}_{\text{TENT}}$	37.26	0.3303	0.0512	3.2092

MIRA has the highest accuracy and lowest NLL in these comparisons and the lowest ECE on CIFAR-100-C. LLLA and ICLA have lower ImageNet-C ECE, but lower accuracy and higher NLL. The resulting trade-offs motivate joint reporting of prediction quality and confidence, rather than ranking methods by MMC or ECE alone.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Introduction state the local stability interpretation and method-dependent gains; Sections 5 and 6 distinguish controlled, repeated, and single-run evidence.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 discusses posterior interpretation, feature-shift failure, hyperparameter choices, and incomplete repeated evaluations.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Appendix A states the fixed-feature Gaussian assumptions and derives the pairwise margin probability, local MI approximation, and exponential weights. No correctness or global optimality guarantee is claimed.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Appendix D reports the available settings and protocol distinctions. Exact interpolation/fallback and some extension configuration details are not fully specified in the available manuscript.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: Existing benchmarks and released baseline implementations are identified, but a public de-anonymized MIRA code release with complete reproduction instructions is not linked here.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [No]

   Justification: The main setup and Appendix D specify available optimizers, scale construction, and defaults; some extension and interpolation details remain incomplete.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Table 4, Appendix D.8, and the ImageNet-Sketch comparison report mean and sample SD over three runs. Accuracy SD is in percentage points; single-run results and small differences are explicitly identified.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: Appendices B–D identify the GPU and report runtime/peak memory for cost comparisons, but total project compute and complete per-experiment resource accounting are unavailable.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: Yes.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Section 6 discusses confidence reliability, error reinforcement, population-dependent effects, and downstream validation needs.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: N/A

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [No]

    Justification: The paper identifies existing benchmarks and released baselines, but complete license and version documentation for every reused asset is not provided.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.

- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: N/A

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: N/A

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: N/A

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: N/A

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.